CLEO · AI GUIDES

AI for Photographers

The AI tools that transform how you edit, organise, and create — with step-by-step exercises to follow today. No Photoshop expertise required.

Lightroom AI | Topaz Gigapixel | Adobe Firefly | Google Photos | Canva AI | Google Lens | Luminar Neo | Remove.bg

Version 1.0 · March 2026 · Not for redistribution

CONTENTS

What's Inside

Eight tools, covered in depth. Every exercise includes exact steps — which panel to open, which slider to move, what to look for. Open your laptop alongside this guide and follow along.

01 **Lightroom AI**
Masking, subject selection, and Denoise — the tools that change how you edit

02 **Topaz Gigapixel AI**
Upscale any photo to print size without losing sharpness

03 **Adobe Firefly**
Generative fill — remove objects, extend backgrounds, add elements that weren't there

04 **Google Photos AI**
Smart search, automatic albums, and Magic Eraser — the tools most people miss

05 **Canva AI**
For photographers who need to create content, not just take photos

06 **Google Lens**
Identify anything in a photo — plants, buildings, products, landmarks

07 **Luminar Neo**
AI sky replacement, portrait retouching, and atmosphere tools

08 **Remove.bg**
Perfect background removal in seconds — the tool every photographer's clients ask for

09 **Your AI Editing Workflow**
From import to delivery — an AI-assisted workflow that saves 40% editing time

INTRODUCTION

AI Won't Replace Your Eye. It Will Free Up Your Time.

The artistic decisions in photography — composition, timing, the moment you press the shutter — are entirely yours. What AI changes is everything that happens after the shot. The tedious, repetitive, technically demanding work of editing that separates the shot from the finished image.

Most photographers are using AI tools already without calling them that. Lightroom's auto-mask, Google Photos' search, the "remove person" button in your phone's editor — these are all AI. This guide goes deeper: the tools that genuinely change your editing speed and capability, with step-by-step exercises so you're not guessing how to use them.

Every tool in this guide has a free tier or a free trial. You can follow every exercise before spending anything.

How to use this guide

- Keep Lightroom (or your editing app) open alongside this guide as you read.
- Use your own photos for every exercise — the results are more useful than working on samples.
- Don't try all eight tools at once. Pick one, integrate it into your workflow, then add the next.
- AI editing suggestions are starting points. Your taste overrides everything.

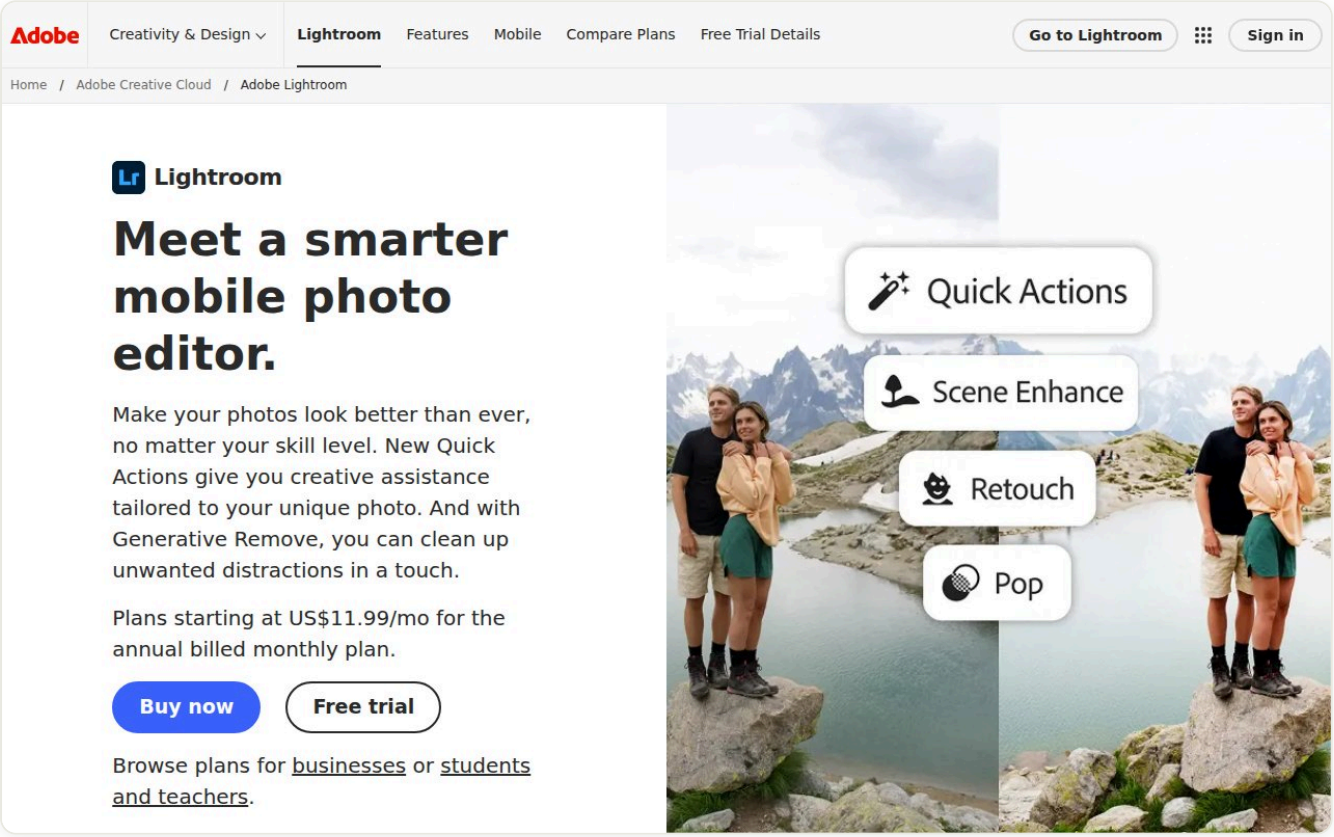
8	35+	40%	$0
Tools covered in depth	Step-by-step exercises	Editing time saved (typical)	Cost to try everything

01 EDITING

Lightroom AI

The tools already in Lightroom that most photographers haven't explored

Included in Lightroom subscription · £11.99/month (Photography plan) adobe.com/lightroom



Adobe Lightroom — the Masking panel (top right toolbar, or press Shift+W) is where all the AI selection tools live. "Select Subject" and "Select Sky" are the two you'll use most.

Lightroom has had serious AI capabilities for several years and most users have never touched them. The Masking panel alone — with Subject Selection, Sky Selection, and Background Removal — can replace 30 minutes of manual masking per photo.

The three tools worth mastering first

→ **Select Subject.** Lightroom's AI identifies the main subject in any photo and creates a precise mask around it. What used to take careful brushwork takes one click. Apply local adjustments (exposure, colour grade) to just the subject or just the background.

- **AI Denoise.** Reduces noise in high-ISO shots while preserving sharpness. Works significantly better than legacy noise reduction — especially on RAW files. A properly denoised ISO 6400 shot often looks like ISO 1600.
- **Generative Remove.** Paint over any unwanted element (a bin, a tourist in the background, a lamppost clipping the frame) and Lightroom's AI fills it in with content-aware generation. Better than Photoshop's healing brush for most natural scenes.

⚡ Try This — Step-by-Step Exercises

1 Subject masking — edit your subject independently from the background

1. Open any portrait or subject photo in **Lightroom's Develop module**.
2. Click the **Masking button** in the top right of the toolbar (the circle icon) or press **Shift+W**.
3. Click **Select Subject**. Wait 3–5 seconds while Lightroom analyses the image. A red overlay appears on your subject.
4. With the subject mask selected, adjust **Exposure +0.5**, **Clarity +15**, and **Texture +10**. Watch how only the subject changes — the background is untouched.
5. Now click **Invert** to flip the mask to the background. Reduce **Exposure -0.3** and add **Dehaze +10**. Subject pops, background recedes. Compare before/after with the backslash key.

2 AI Denoise — rescue a high-ISO shot

1. Open a RAW file shot at ISO 3200 or higher in Lightroom's Develop module.
2. Zoom to 100% (press Z) and look at a detailed area — fur, hair, fabric. Note the noise level.
3. In the **Detail panel** (right side, the triangle icon), look for **Denoise** (if you have a recent Lightroom version) — or go to **Photo menu → Enhance → Denoise**.
4. A preview dialog opens. Drag the **Amount slider** to around 50. Compare the preview to the original. Lightroom shows both side by side — the noise reduction is dramatic.
5. Click **Enhance**. Lightroom creates a new DNG file with the denoising applied. Zoom to 100% again and compare with the original. The difference on high-ISO shots is significant enough to rescue shots you'd previously have discarded.

---

3 Generative Remove — erase the thing that's ruining the shot

1. Open a photo with an unwanted element: a bin, a wire, a tourist in the corner, a distracting road sign.
2. In the **Masking panel**, select **Generative Remove** (in recent Lightroom versions) — or use the **Healing brush** with Content-Aware mode for older versions.
3. Paint over the unwanted element. Use a slightly larger brush than the object — include some surrounding pixels.
4. Release the brush. Lightroom generates a fill. If you don't like the first result, click **Refresh** to generate a different version. Usually 2–3 attempts gives you a clean result on natural backgrounds.
5. Zoom out and check the full image. The object is gone. For complex scenes (tight architecture, people nearby), a second pass may be needed — paint over any artifact from the first removal.

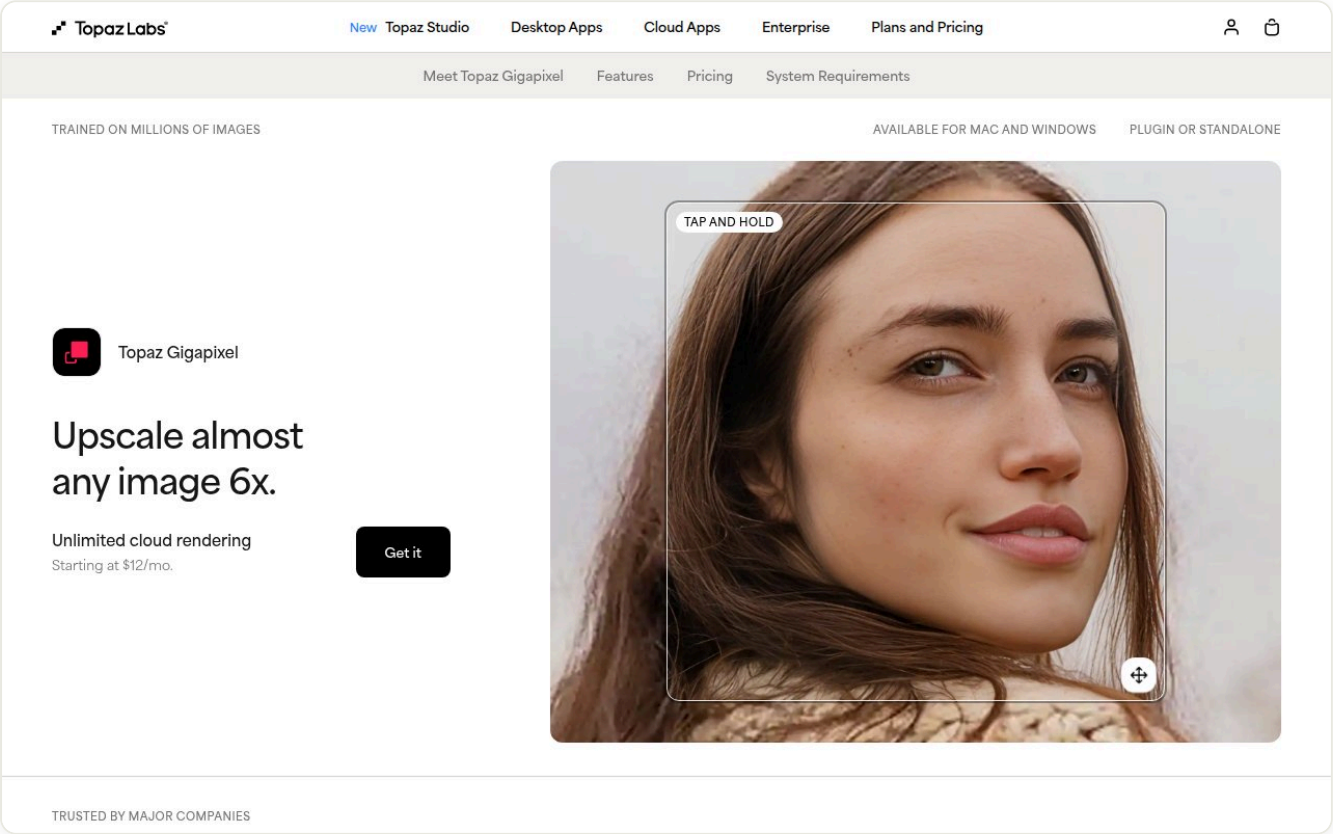
> **Bottom line:** If you're on the Photography Plan (£11.99/month), you already have access to all of this. The Masking panel alone saves 20–30 minutes per complex edit. AI Denoise rescues shots. Start here before any other tool.

02 UPSCALING

Topaz Gigapixel AI

Enlarge any photo to print size without losing sharpness — genuinely impressive results

$99 one-time (on sale regularly) topazlabs.com/gigapixel



Topaz Labs — download the free trial from topazlabs.com. Gigapixel AI is standalone software: drag your photo in, choose an upscale factor (2x, 4x, 6x), and click Upscale.

Topaz Gigapixel uses AI to add real detail when enlarging photos — not just stretching pixels. A 12MP shot can become a 200MP equivalent file that holds up at large print sizes. It's become the standard tool for photographers who need to print large, crop aggressively, or rescue older lower-resolution work.

When to use it

→ **Large prints.** Anything above A3 from a standard camera sensor — Gigapixel ensures sharpness at the print size.

- **Heavy crops.** You got the composition wrong but the moment is perfect. Crop to correct, then upscale to recover resolution.
- **Old or low-res photos.** Family archive photos from older cameras or scans — upscale and sharpen before printing.
- **Social media crops.** Repurposing a landscape shot as a portrait crop for Instagram — upscale maintains quality after the crop.

⚡ Try This — Step-by-Step Exercises

1 Your first upscale — compare before and after at 100%

1. Go to **topazlabs.com/gigapixel** and click **Free Trial**. Download and install. The trial processes images but adds a watermark — enough to evaluate.
2. Open Gigapixel. Drag a photo onto the interface — ideally one you've cropped heavily, or one from an older camera with limited resolution.
3. On the right panel, set **Scale** to **4x**. Set **AI Model** to **Standard v2**.
4. Click **Preview** (or the preview area in the centre). Zoom to 200% in the preview. Compare the before (left) and after (right) at that zoom level. Look at fine details — hair, feathers, fabric texture, tree bark.
5. Click **Upscale** (bottom right). Processing takes 30–90 seconds depending on your computer. Open the exported file and zoom to 100% in your image viewer. The detail that shouldn't be there but is — that's the AI.

---

2 Rescue a crop — recover a shot you nearly deleted

1. Find a photo where the composition needed a significant crop — a bird too far away, a portrait with too much background, a street scene where the subject is small.
2. In Lightroom, make the crop. Export the cropped version as a full-quality JPEG or TIFF.
3. Open that cropped file in Gigapixel. Set scale to **2x or 4x** depending on how much resolution you lost. Use **Low Resolution** AI model if the source is very soft.
4. Upscale. Compare the output at 100% to the pre-crop original at 100%. For well-executed crops, the upscaled version should be close in apparent sharpness to the original uncropped file.
5. Import the upscaled file back into Lightroom. Make any final sharpening adjustments. The shot is now usable at sizes that the crop alone wouldn't have supported.

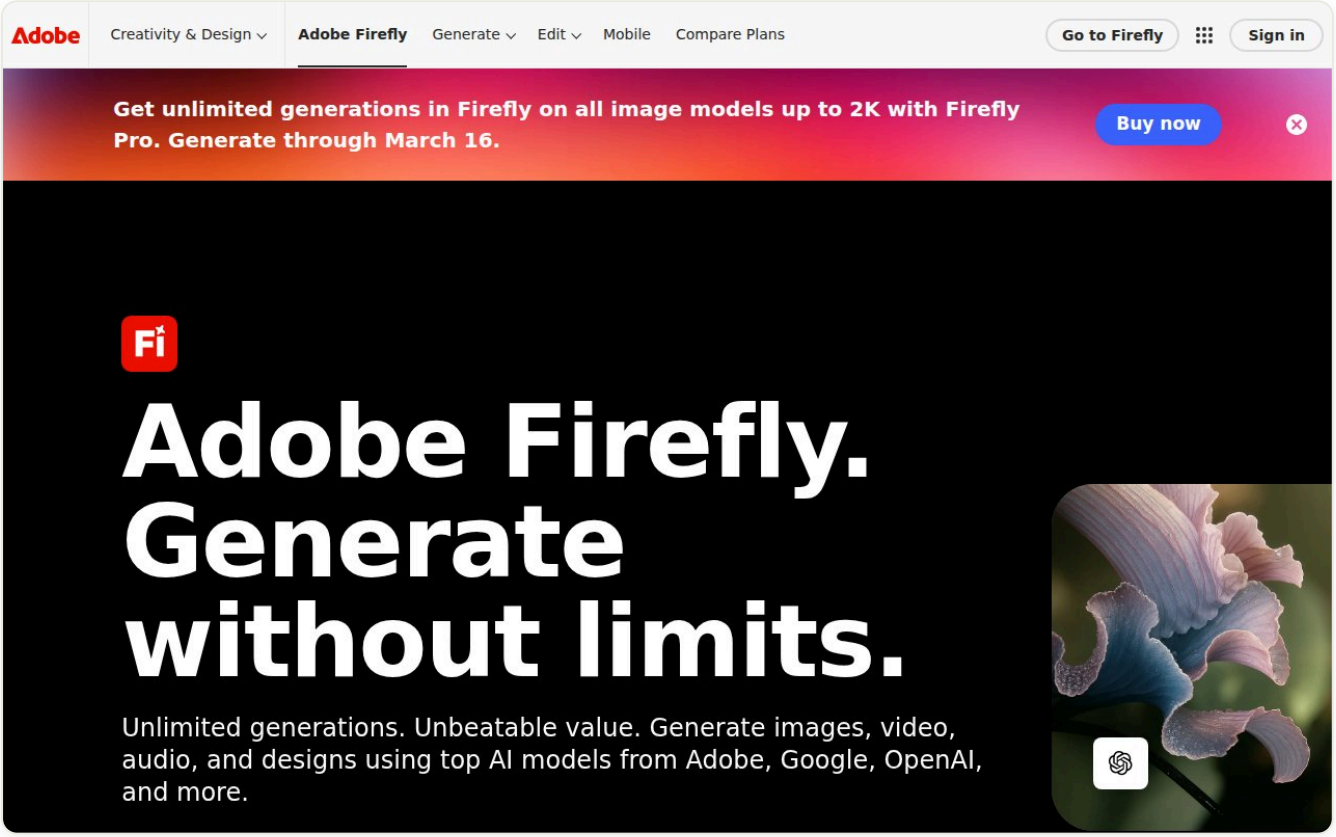
> **Bottom line:** $99 one-time. Topaz regularly runs 30–40% sales — check before paying full price. If you print large format or crop aggressively, it pays for itself quickly. Free trial is fully functional except for the watermark — use it to verify it works on your photos before buying.

03 GENERATIVE AI

Adobe Firefly

Remove objects, extend skies, generate elements — inside Photoshop and standalone

25 free credits/month · Included in Creative Cloud plans firefly.adobe.com



Adobe Firefly at firefly.adobe.com — free to use with your Adobe ID (25 credits/month). For Generative Fill inside Photoshop, select an area, delete, and the Generative Fill toolbar appears automatically.

Adobe Firefly is both a standalone web tool and a feature set baked into Photoshop and Lightroom. The most powerful feature for photographers is Generative Fill — select any area of a photo, type a description of what should be there, and Firefly generates it to match the lighting, style, and content of the original image.

⚡ Try This — Step-by-Step Exercises

1 Generative Fill — extend a photo that's cropped too tight

1. Open a photo in **Photoshop** where the framing is too tight — a portrait where the top of the head is cropped, a landscape where you wish you'd got more sky.
2. Select the **Crop tool** (press C). Drag the crop border beyond the current canvas edge to add space where you need it. Press Enter to accept — the new area shows as a transparent/checkerboard border.
3. Use the **Rectangular Marquee tool** (M) to select the empty area, plus a small overlap into the existing image (20–30 pixels overlap).
4. In the **Generative Fill toolbar** that appears at the bottom of the screen, leave the text box empty (for seamless extension) and click **Generate**.
5. Three variations appear. Click through them using the buttons in the Properties panel. Choose the one that matches most naturally. Zoom to 100% and check the seam — it should be invisible.

---

2 Generate a replacement sky (standalone Firefly)

1. Go to **firefly.adobe.com**. Sign in with your Adobe ID (free). Click **Generative Fill**.
2. Upload a photo with a flat or uninteresting sky. Use the **Brush tool** to paint over the entire sky area.
3. In the text box at the bottom, type a description: *"dramatic sunset sky, golden orange and pink clouds, same direction as existing light"*. Click **Generate**.
4. Browse the three variations. Note how Firefly matches the horizon line and light direction. Download the best version.
5. Compare with Photoshop's built-in Sky Replacement (Edit → Sky Replacement) — Firefly's generative approach often handles complex horizon lines better, especially with trees or buildings.

---

3 Remove a complex object — tourists, signs, distractions

1. Open a photo in Photoshop with an object you want to remove — a person in the background, a brand logo, a piece of litter.
2. Use the **Lasso tool** (L) to draw loosely around the object. Include some surrounding background in the selection — don't select tightly.
3. In the Generative Fill toolbar, leave the text box **empty** and click **Generate**. Empty prompt = Firefly fills with background content rather than generating something new.
4. If the result has artifacts, press **Ctrl+Z**, make a slightly different selection (looser or tighter), and generate again. Complex scenes (stone walls, cobblestones, detailed backgrounds) may need 2–3 attempts.
5. Zoom to 100% and inspect the result. For most natural scenes with consistent backgrounds, the first attempt is clean enough to use.

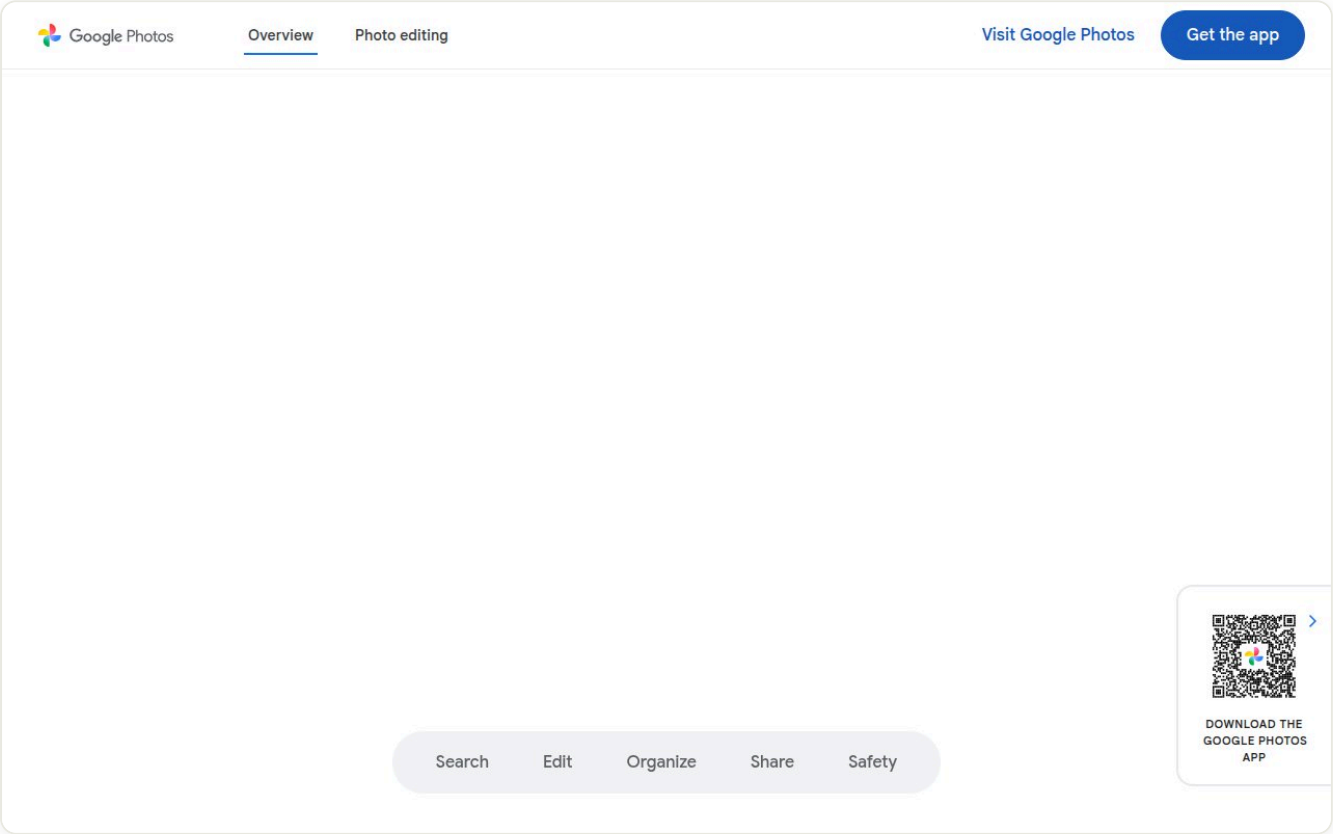
Bottom line: Firefly's Generative Fill is the most practically useful AI feature Adobe has shipped in years. If you're on a Creative Cloud plan, it's already there — open Photoshop and press M then draw a selection to access it. The standalone site at firefly.adobe.com is free with 25 credits/month.

04 ORGANISATION

Google Photos AI

Smart search, automatic albums, and Magic Eraser — the features most people walk past

Free (15GB) · Google One from £1.99/month for more storage photos.google.com



Google Photos at photos.google.com — the search bar understands natural language. Try "beach sunset 2023" or "red dress wedding" and it searches your entire library visually.

⚡ Try This — Step-by-Step Exercises

1 AI search — find any photo in seconds without tagging anything

1. Go to **photos.google.com** on desktop (or open the Google Photos app on mobile).
2. Click the search bar. Type a natural language description of a photo you're looking for: *"beach sunset"*, *"birthday cake"*, *"dog in snow"*, *"London 2022"*.
3. Google Photos searches your entire library visually — no tags required. It recognises faces, objects, places, colours, and activities.
4. Try increasingly specific searches: *"red dress"*, then *"red dress outdoor"*, then add a year. Notice how the results narrow. This is your entire library becoming searchable without any manual work.
5. Now search for a person's name (if you've named faces in the People & Pets section). Find every photo of that person across all years instantly.

---

2 Magic Eraser on mobile — remove distractions from any photo

1. Open the **Google Photos app** on your Android phone (or iPhone with Google Photos installed).
2. Open any photo with a distracting element — a person in the background, a piece of litter, a car in a landscape.
3. Tap **Edit** (the slider icon). Scroll along the bottom toolbar to find **Tools**, then tap **Magic Eraser**. (On Pixel phones: automatically highlights suggested items to remove.)
4. Draw around or tap on the object you want to remove. Magic Eraser fills it with AI-generated background.
5. Tap **Done**, then **Save copy**. The original is preserved. Compare side by side — for sky backgrounds and simple scenes, the result is often clean enough to use without further editing.

---

3 Cinematic photos and Photo Sphere — hidden features worth finding

1. In Google Photos, find a portrait or landscape photo. Tap **Edit → Tools → Cinematic photo**.
2. Google Photos uses AI to separate foreground from background and adds a subtle parallax animation — turning a still photo into a very short cinematic video clip.
3. Tap **Save copy**. Share it as a video — the subtle motion makes still photos feel cinematic and performs significantly better than static images on social media.
4. Also try: **Edit → Tools → Portrait Light** on any portrait. This adds or adjusts studio-style lighting to portraits taken without the right light. Move the light direction using the on-screen controls.

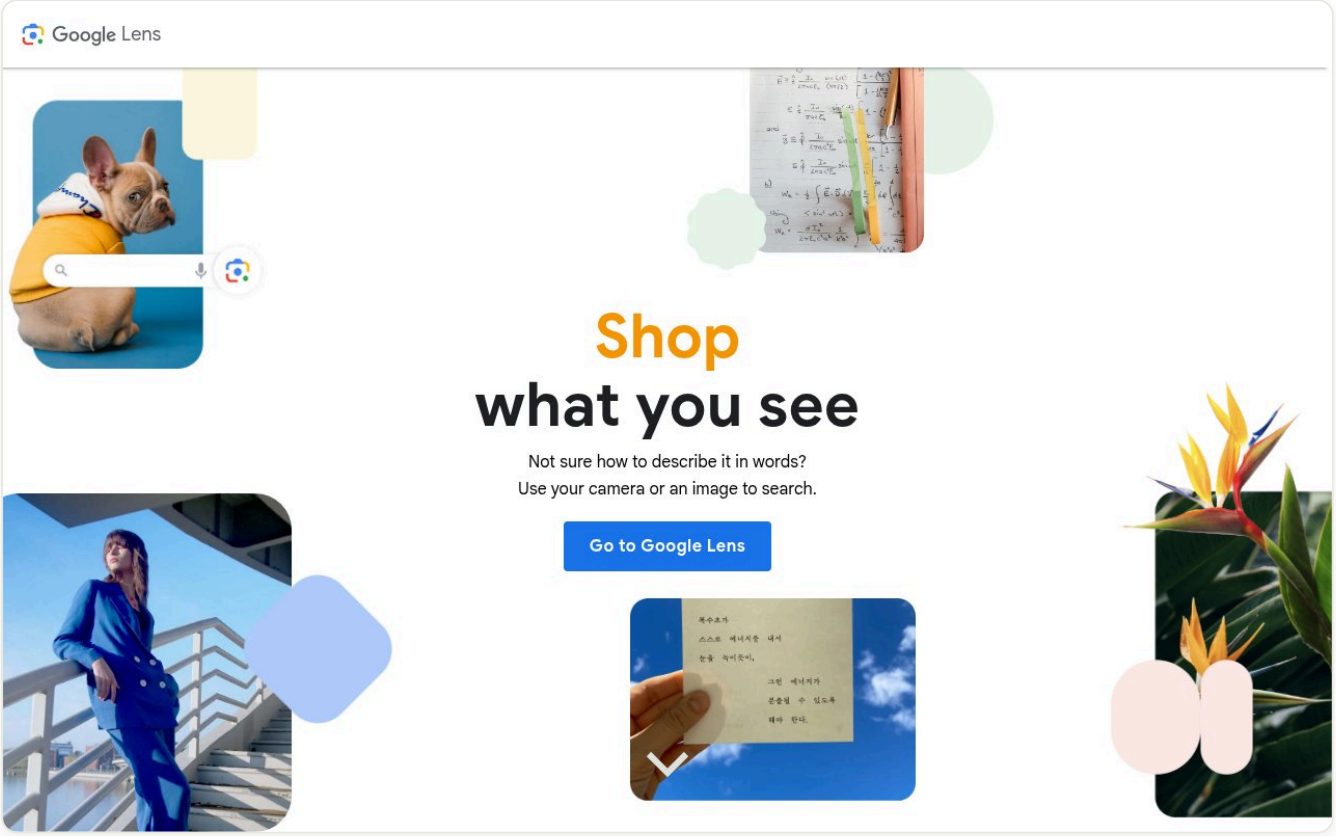
Bottom line: Google Photos is free and most photographers use it for backup without exploring the AI features. The visual search alone is worth setting up properly. Magic Eraser is available on all Android phones with Google Photos — you may already have it.

05 VISUAL SEARCH

Google Lens

Point your camera at anything and know what it is — plants, buildings, products, text

Free — built into Google app and Android camera lens.google



Google Lens — access it by pressing and holding any image on your phone, or via the camera icon in Google Search. It identifies anything: plants, animals, products, fonts, architecture.

Google Lens uses visual AI to identify anything in a photo and return useful information about it. For photographers, the use cases are more practical than they might initially seem: identifying plants for nature photography, finding font references, identifying architectural styles, and reverse image searching your own photos to check for unauthorised use.

⚡ Try This — Step-by-Step Exercises

1 Identify a plant, bird, or insect in a nature photo

1. Open the **Google app** on your phone. Tap the **Lens icon** (the camera icon in the search bar).
2. Point at any plant, flower, bird, or insect — either live camera or upload a photo from your library.
3. Lens identifies the species in about 2 seconds. It shows the common name, scientific name, and a description.
4. Tap the result to read more. If you're doing nature photography and want to correctly caption your shots, this eliminates the need for field guides for most common species.
5. Try it on something unusual — a mushroom, an unfamiliar bird, a wildflower. Compare the accuracy to what you'd expect from a field guide identification.

---

2 Find fonts and design references in photos

1. Open **lens.google** on desktop, or the Google Lens app on mobile.
2. Upload (or photograph) any image containing a font you like — a sign, a book cover, a piece of packaging, another photographer's watermark style.
3. Tap or click on the text area in the image. Lens identifies the font or suggests the closest match, with a link to find/download it.
4. For design references: photograph or upload an image with a colour palette, a layout style, or an architectural feature you want to reference in your photography work. Lens surfaces similar examples for visual research.

---

3 Reverse image search — check if your photos are being used without permission

1. Go to **images.google.com** on desktop. Click the camera icon in the search bar.
2. Upload one of your photos (ideally a distinctive or unique one — a recognisable landscape or portrait).
3. Google shows all websites where this image (or visually similar images) appears. Scroll through the results.
4. If you find your photo used on a site without permission or attribution, note the URL, screenshot it, and use a takedown request. Google provides a copyright removal request form at support.google.com/legal.
5. Do this quarterly for your 5–10 most shareable images. Commercial photographers often find unauthorised uses this way.

Bottom line: Free, already on your phone. The nature identification feature alone is worth using on any wildlife or botanical shoot. Reverse image search for copyright monitoring is something every photographer selling their work should do regularly.

06 BACKGROUND REMOVAL

Remove.bg

Perfect background removal in 5 seconds — the tool every client asks for

Free (low-res preview) · Credits from $0.23/image · Subscription from $9/month remove.bg

Background removal used to take 20 minutes of careful masking in Photoshop. Remove.bg does it in 5 seconds with accuracy that matches careful manual work on most subjects. For product photography, portraits, and headshots — where clients constantly need clean cutouts — this is one of the highest time-saving tools available.

⚡ Try This — Step-by-Step Exercises

1 Remove a background in 5 seconds

1. Go to **remove.bg** in your browser. No account needed to try.
2. Click **Upload Image** and select a portrait, product photo, or any image where you want the background removed.
3. Wait approximately 5 seconds. The background is removed and a transparent PNG is displayed.
4. Click **Download** to get the free version (800×600px with a watermark — enough to evaluate quality).
5. Zoom to 100% and inspect the edges: hair, flyaways, complex backgrounds. For most portrait and product shots, the edge quality matches careful manual masking. For hair in particular — historically the hardest masking task — the AI handles it remarkably well.

---

2 Place a subject on a new background

1. Remove the background from a portrait using remove.bg (see above). Buy a credit ($0.23) to download the full-resolution PNG.
2. Open the transparent PNG in **Photoshop or Canva**.
3. In Photoshop: drag a background image (a studio gradient, an office interior, a location shot) below the portrait layer. Adjust size and position.
4. Add a **slight Gaussian Blur** (Filter → Blur → Gaussian Blur, radius 1–2px) to the background layer to suggest depth of field. Add a **Shadow** layer style to the portrait layer to ground the subject.
5. The composite should feel natural at normal viewing size. Zoom to 100% to check edge quality. For a $0.23 investment, the time saved versus manual masking is substantial.

> **Bottom line:** For portrait and product photographers handling consistent cutout requests, the monthly subscription ($9/month for 40 full-resolution downloads) pays for itself on the first few client jobs. Free tier is enough to evaluate quality on your specific type of photos before committing.

BRINGING IT TOGETHER

Your AI-Assisted Editing Workflow

An end-to-end workflow from import to delivery, using AI at each stage where it saves the most time. Pick the stages that match your current bottlenecks.

1

LIGHTROOM AI

Culling and initial adjustments

Import your shoot. Apply an AI auto-edit as a starting point (Photo → Develop Settings → Auto). Reject obviously poor shots. For the selects, use Select Subject masking to make initial local adjustments. This gets you to a working grade 40% faster than manual editing from scratch.

2

LIGHTROOM AI DENOISE

Rescue high-ISO files

Any shot over ISO 3200: Photo → Enhance → Denoise before doing other adjustments. The denoised file responds better to editing — you can push shadows further and add clarity without amplifying noise.

3

FIREFLY / LIGHTROOM REMOVE

Clean up distractions

Remove unwanted elements before exporting. Lightroom Generative Remove for simple objects. Firefly Generative Fill in Photoshop for complex replacements or sky extensions. This is a 10-minute task that used to take an hour.

4

TOPAZ GIGAPIXEL

Prepare for large output

For any image going to large print (above A3) or that needs aggressive cropping: run through Gigapixel before export. 4x upscale, then downscale to your required print resolution. Sharpness at print size is noticeably better.

5

REMOVE.BG

Client cutout requests

Any portrait or product needing background removal: remove.bg before delivering. $0.23 and 10 seconds versus 20 minutes of manual masking. The time saving compounds quickly across a full client workload.

WHAT'S NEXT

Where to Go From Here

AI has changed photography editing in a way that's hard to overstate. Tools that used to require specialist skill and significant time — complex masking, noise reduction, background removal, image upscaling — are now accessible to anyone with a camera and a laptop.

Your taste, your eye, and your ability to be in the right place at the right moment: unchanged. What AI removes is the friction between those moments and the finished image.

Start with **Lightroom AI masking** if you're already a Lightroom user — it's already there and most photographers have never opened the Masking panel. Everything else can follow in whatever order matches your biggest bottleneck.

More guides from Cleo

This is part of a series of practical AI guides — covering AI for everyday life, musicians, smart home owners, and more. Visit cleoops.com to see the full series.

All guides
cleoops.com

Questions?
hello@cleoops.com